What is the reported malignancy prevalence in surgical cohorts using ACR TI-RADS classification, and how do authors address potential surgical selection bias? "Find studies evaluating ACR TI-RADS malignancy rates in surgical cohorts (not screening cohorts). Identify reported malignancy prevalence, whether sensitivity/specificity were reported, and whether authors acknowledged surgical selection bias."

Reported malignancy prevalence in surgical cohorts using ACR TI-RADS ranged widely from 12% to 66% depending on patient selection criteria, with the majority of studies reporting diagnostic performance metrics, but only 8% of studies explicitly acknowledged surgical selection bias despite the substantially inflated malignancy rates compared to screening populations.

Abstract

This systematic review of 25 studies evaluating ACR TI-RADS in surgical cohorts found substantial heterogeneity in reported malignancy prevalence, ranging from 12.2% to 66.1% . Studies with restrictive inclusion criteria (e.g., indeterminate cytology only) reported higher malignancy rates (47-66%) , while prospective consecutive enrollment studies showed the lowest rates (12.2-12.5%) , most closely approximating screening population expectations. Category-specific malignancy rates demonstrated consistent gradients across large studies, with TR5 rates of 67-100% and TR2-3 rates of 0-13% , though these substantially exceeded ACR TI-RADS thresholds established for screening populations. The majority of studies (23/25) reported sensitivity and specificity, with sensitivity generally ranging from 51.6% to 100% and specificity from 38.1% to 92.77% , depending on the threshold used for positive classification.

Despite the inherent selection bias in surgical cohorts, only 2 of 25 studies (8%) explicitly acknowledged this limitation , with 5 additional studies (20%) providing partial acknowledgment through discussion of generalizability concerns . The remaining 72% of studies did not address how surgical selection affects malignancy prevalence estimates or whether results apply to screening populations. This gap is clinically significant, as the inflated malignancy rates observed in surgical cohorts reflect appropriate clinical decision-making based on factors beyond ultrasound appearance alone, and should not be extrapolated to estimate risk in unselected populations undergoing thyroid nodule screening.

Paper search

We performed a semantic search using the query "What is the reported malignancy prevalence in surgical cohorts using ACR TI-RADS classification, and how do authors address potential surgical selection bias? "Find studies evaluating ACR TI-RADS malignancy rates in surgical cohorts (not screening cohorts). Identify reported malignancy prevalence, whether sensitivity/specificity were reported, and whether authors acknowledged surgical selection bias."" across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We retrieved the 499 papers most relevant to the query.

Screening

We screened in sources based on their abstracts that met these criteria:

- **ACR TI-RADS Classification**: Does this study evaluate thyroid nodules using the ACR TI-RADS classification system?
- **Surgical Cohort with Histopathology**: Does this study include surgical cohorts (patients who underwent thyroid surgery) with histopathological confirmation as the reference standard?
- **Malignancy Prevalence Data**: Does this study report malignancy prevalence rates with sufficient data that can be extracted for analysis?
- **Appropriate Study Design**: Is this study an original research study (prospective cohort, retrospective cohort, or cross-sectional study) or a systematic review/meta-analysis?
- **Adequate Sample Size**: Does this study include 10 or more patients (i.e., is it NOT a case report or case series with fewer than 10 patients)?
- **Adult Population**: Does this study focus on adult populations (i.e., is it NOT exclusively focused on pediatric populations)?
- **Full Publication with Methodological Detail**: Is this a full-text publication with sufficient methodological detail (i.e., is it NOT a conference abstract, editorial, letter, or commentary)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Study Population**:

  Extract details about the surgical cohort including:

- Total number of patients and nodules
- Patient demographics (age, gender distribution)
- Study timeframe and setting
- Inclusion/exclusion criteria for surgery
- Types of thyroid surgery performed
- Whether this was a consecutive surgical series or selected cases

- **Malignancy Prevalence**:

  Extract malignancy rates by ACR TI-RADS category:

- Overall malignancy rate in the cohort (percentage)
- Malignancy rate for each TI-RADS category (1, 2, 3, 4, 5)
- Number of nodules in each category
- How malignancy was confirmed (histopathology)
- Any subgroup analyses (e.g., by nodule size, cytology results)

- **Diagnostic Performance**:

  Extract reported diagnostic accuracy metrics:

- Sensitivity and specificity (with confidence intervals if provided)
- Positive and negative predictive values
- Area under the curve (AUC)
- Which TI-RADS categories were considered 'positive' for analysis

- Any comparison with other classification systems
- Performance in different nodule size groups if reported

- **Selection Bias Discussion**:

  Extract how authors address surgical selection bias:

- Whether authors explicitly acknowledge that surgical cohorts have selection bias
- Any discussion of how surgical selection affects malignancy prevalence estimates
- Whether authors mention that their results may not apply to screening populations
- Any attempts to account for or adjust for selection bias
- Discussion of study limitations related to surgical selection
- Whether authors recommend caution in generalizing results

Characteristics of Included Studies

This systematic review examines 25 studies evaluating ACR TI-RADS malignancy rates in surgical cohorts. The studies span multiple geographic regions and healthcare settings, with varying sample sizes and study designs.

Study	Full text retrieved?	Study Type	Sample Size (Patients/Nodules)	Study Period	Setting
I. Ahmadinejad et al., 2023	No	Retrospective	146 patients	2018-2020	Not specified
Thayse Lozovoy Madsen Barbosa et al., 2019	Yes	Retrospective	139 patients, 140 nodules	2012-2016	Brazil, two institutions
Orhan Asya et al., 2022	No	Retrospective	120 patients	2017-2019	Single center
S. Ahmadi et al., 2019	No	Retrospective	186 patients, 202 nodules	2010-2017	Quaternary hospital
Wei-bing Zhang et al., 2020	No	Retrospective	1271 nodules	Not specified	Not specified
M. D. de Jong et al., 2022	No	Retrospective	214 patients	2018-2020	Not specified
S. Samargandy et al., 2024	Yes	Retrospective	191 patients	2017-2022	King Abdulaziz University
Ahmed Alsibani et al., 2024	No	Retrospective	329 patients, 381 nodules	2021-2023	Not specified
Luying Gao et al., 2019	No	Retrospective	1758 patients, 2544 nodules	2015	Single center
S. Ahmadi et al., 2019a	No	Retrospective	213 patients, 323 nodules	2016	Tertiary care hospital
K. Daniels et al., 2020	No	Retrospective	121 patients, 136 nodules	2017-2018	Not specified
Yi Zheng et al., 2018	No	Retrospective	1013 patients, 1033 nodules	2015-2016	Single hospital

Study	Full text retrieved?	Study Type	Sample Size (Patients/Nodules)	Study Period	Setting
Yun Zhu et al., 2020	No	Retrospective	2399 patients, 2643 nodules (cohort 1); 1837 patients, 1905 nodules (cohort 2)	2010-2014; 2017-2018	Jiangsu Institute of Nuclear Medicine
Vinay Raj Thattarakkal et al., 2021	No	Prospective	85 patients	1 year	Tertiary care hospital
Saad M. Alqahtani et al., 2022	No	Retrospective	110 cases	4 years	Not specified
Zeyad T. Sahli et al., 2019	Yes	Retrospective	131 patients, 133 nodules	2012-2016	Johns Hopkins Hospital
Chi Fung Chia et al., 2023	No	Retrospective	164 cases	2021-2022	Regional hospital, Hong Kong
R. E. Khalushi et al., 2020	No	Retrospective	104 patients	2001-2015	South Africa, two hospitals
Bengi Balci et al., 2023	Yes	Prospective	62 patients, 151 nodules	2017-2018	Single tertiary center
Kyla Wright et al., 2022	No	Retrospective	496 nodules	2018-2021	Academic health system
K. Anwar et al., 2023	Yes	Prospective	205 patients	2019-2022	Hayatabad Medical Complex, Peshawar
M. R. Romero Ibarguengoitia et al., 2024	No	Retrospective	160 patients	2022-2023	4 centers, Northern Mexico
Soumya Sarayu et al., 2025	No	Prospective	295 nodules with definitive outcome	Not specified	Not specified
M. Paker et al., 2021	No	Retrospective	265 patients, 238 nodules	2012-2019	Single hospital

The majority of included studies (21/25) employed retrospective designs, while four studies were prospective . Sample sizes varied considerably, ranging from 62 patients to 2,399 patients . Female predominance was consistently reported across studies, ranging from 71.4% to 93.3% . Mean patient ages generally ranged from 37.68 years to 57 years .

Malignancy Prevalence

Overall Malignancy Rates

The overall malignancy prevalence in surgical cohorts showed substantial heterogeneity across studies, reflecting differences in patient selection, institutional practices, and referral patterns.

Study	Overall Malignancy Rate	Cohort Characteristics
Luying Gao et al., 2019	66.1%	Large surgical cohort, single center
Orhan Asya et al., 2022	50%	Patients scheduled for surgery
M. Paker et al., 2021	48.3%	Thyroidectomy patients
K. Daniels et al., 2020	47.1%	Indeterminate cytology cohort
S. Samargandy et al., 2024	47.6%	Non-benign FNA or compressive symptoms
Saad M. Alqahtani et al., 2022	47.3%	AUS/FLUS cytology only
Thayse Lozovoy Madsen Barbosa et al., 2019	47.1%	Indeterminate cytology (Bethesda III-V)
Yi Zheng et al., 2018	29.8%	Mixed surgical and FNA cohort
S. Ahmadi et al., 2019a	27.2%	Nodules >5mm undergoing thyroidectomy
Chi Fung Chia et al., 2023	26.8%	Thyroidectomy cases
Zeyad T. Sahli et al., 2019	22.6%	Indeterminate, Afirma-suspicious nodules
R. E. Khalushi et al., 2020	22.8%	Nodular thyroid goiter with surgery
K. Anwar et al., 2023	12.20%	Mixed solitary and multinodular goiters
Soumya Sarayu et al., 2025	12.5%	Prospective consecutive subjects
Kyla Wright et al., 2022	12.3%	TI-RADS category 5 nodules only

The wide range of malignancy prevalence (12.2% to 66.1%) reflects the substantial impact of patient selection criteria on study outcomes. Studies focusing exclusively on indeterminate cytology showed higher malignancy rates (47.1-47.3%) , while those including broader populations or employing prospective consecutive enrollment demonstrated lower rates (12.2-12.5%) .

Malignancy Rates by ACR TI-RADS Category

The malignancy rates by TI-RADS category generally demonstrated an increasing gradient with higher risk categories, though absolute values varied substantially between studies.

Study	TR1	TR2	TR3	TR4	TR5
Luying Gao et al., 2019	0%	1.3%	9.1%	52.5%	88.8%
Yi Zheng et al., 2018	-	0%	1.1%	13.0%	67.1%

Study	TR1	TR2	TR3	TR4	TR5
Vinay Raj Thattarakkal et al., 2021	-	4.2%	13.3%	57.9%	100%
M. Paker et al., 2021	0%	7.5%	11.4%	59.6%	90.0%
Bengi Balci et al., 2023	4.76%	23.5%	32.6%	40%	90%
S. Samargandy et al., 2024	-	-	34.1%	42.0%	82.5%
Saad M. Alqahtani et al., 2022	-	-	43.5%	49.4%	40%
M. R. Romero Ibarguengoitia et al., 2024	-	-	36%	59%	80%
Thayse Lozovoy Madsen Barbosa et al., 2019	-	20.0%	23.3%	TR4a: 28.1%, TR4b: 55.6%, TR4c: 65.2%	92.9%
S. Ahmadi et al., 2019a	-	-	10%	38%	-

The malignancy rates for TR5 nodules ranged from 40% to 100% , with most studies reporting rates between 80-92% . Notably, the Alqahtani study focusing on AUS/FLUS nodules showed an unexpectedly low TR5 malignancy rate (40%) , with no significant association between TI-RADS and final pathological analysis in this specific cytological category . Lower-risk categories (TR2-TR3) showed more variable results, with some surgical cohorts demonstrating substantially higher malignancy rates than predicted by ACR TI-RADS, particularly in studies with selective surgical criteria .

Diagnostic Performance

Sensitivity and Specificity

The reported diagnostic performance of ACR TI-RADS varied considerably across studies, influenced by the threshold selected for positive classification and the underlying population characteristics.

Study	Sensitivity	Specificity	PPV	NPV	Categories Considered Positive
Soumya Sarayu et al., 2025	100%	60.5%	-	99.5%	TR4-5
K. Anwar et al., 2023	80%	92.77%	-	-	TR3-5

Study	Sensitivity	Specificity	PPV	NPV	Categories Considered Positive
Bengi Balci et al., 2023	82.5%	57%	64.58%	77.67%	TR3-5
M. Paker et al., 2021	84.3%	53.6%	-	-	Not specified
Orhan Asya et al., 2022	80%	56%	72%	67%	TR3-5
I. Ahmadinejad et al., 2023	74.5-79.9%	76.8%	60.3%	76.8%	TR4-5
Ahmed Alsibani et al., 2024	77.9%	57.3%	73.8%	62.8%	TR4-5
S. Ahmadi et al., 2019a	78.4%	73.2%	52.3%	90%	Not specified
Vinay Raj Thattarakkal et al., 2021	77.8%	89.6%	66.6%	93.8%	Not specified
Chi Fung Chia et al., 2023	72.7%	40.8%	-	-	Not specified
Thayse Lozovoy Madsen Barbosa et al., 2019	69.7%	81.1%	76.7%	75.0%	TR4b-5
R. E. Khalushi et al., 2020	69.5%	61.5%	-	-	Not specified
S. Samargandy et al., 2024	51.6%	90%	82.5%	67.2%	TR5 only
M. R. Romero Ibarguengoitia et al., 2024	TR4: 41%, TR5: 51%	TR4: 50%, TR5: 77%	TR4: 59%, TR5: 80%	TR4: 32%, TR5: 47%	Category-specific
Zeyad T. Sahli et al., 2019	71.4%	38.1%	40.2%	69.6%	TR4-5

Sensitivity generally ranged from 51.6% to 100%, while specificity ranged from 38.1% to 92.77%. The highest sensitivity (100%) was achieved using TR4-5 as positive cutoffs in a prospective study , while the highest specificity (92.77%) was reported using TR3-5 as positive cutoffs . Studies using only TR5 as the positive threshold showed lower sensitivity but higher specificity .

Comparison with Other Classification Systems

Several studies directly compared ACR TI-RADS with the ATA guidelines or other classification systems.

Study	ACR TI-RADS Performance	ATA Performance	Key Finding
Wei-bing Zhang et al., 2020	Higher specificity, lower sensitivity	-	ACR TI-RADS had better AUC in <10mm nodules
Luying Gao et al., 2019	Specificity 79.7%	Sensitivity 95.5%	KWAK-TIRADS and ATA showed better diagnostic efficiency
S. Samargandy et al., 2024	Sensitivity 51.6%, Specificity 90%	Sensitivity 52%, Specificity 80%	ACR TI-RADS had superior specificity
S. Ahmadi et al., 2019a	AUC 0.76	AUC 0.77	Similar diagnostic value (κ=0.93)
Chi Fung Chia et al., 2023	Sensitivity 72.7%, Specificity 40.8%	Sensitivity 81.8%, Specificity 10.8%	ATA more sensitive, ACR more specific
M. Paker et al., 2021	Sensitivity 84.3%, Specificity 53.6%	Sensitivity 93.3%, Specificity 15.5%	Both systems AUC >0.88

The consistent finding across comparative studies was that ACR TI-RADS tends to have higher specificity but lower sensitivity compared to ATA guidelines . The level of agreement between ACR TI-RADS and ATA classification was substantial, with weighted kappa statistics of 0.91-0.93 reported .

Performance in Special Populations

Performance in indeterminate cytology cohorts showed variable results. One study reported particularly poor performance of ACR TI-RADS in cytologically indeterminate and Afirma-suspicious nodules, with sensitivity of 71.4%, specificity of 38.1%, and accuracy of only 50.4% . Inter-reader reliability for TI-RADS categorization in indeterminate nodules was marginal (κ=0.293) , with intra-reader reliability ranging from marginal to good (κ=0.337-0.560) . The AUC for predicting high-risk genetic mutations was only 0.509, suggesting no optimal TI-RADS level for identifying high-risk nodules in this population .

Performance by nodule size was reported in several studies, with ACR TI-RADS and other systems performing better in nodules >1cm compared to smaller nodules . Studies noted potential issues with size thresholds for FNA recommendations, as 10% of TR3 nodules <2.5cm and 38% of TR4 nodules <1.5cm were malignant .

Selection Bias Discussion

Acknowledgment of Selection Bias

The majority of included studies did not explicitly acknowledge or address surgical selection bias.

Study	Explicit Acknowledgment	Discussion of Impact	Recommendations for Caution
Soumya Sarayu et al., 2025	Yes - explicitly acknowledged selection bias in prior studies	Not discussed	Not provided

Study	Explicit Acknowledgment	Discussion of Impact	Recommendations for Caution
Zeyad T. Sahli et al., 2019	Yes - acknowledged retrospective nature and selection bias	Discussed impact of patient selection on results	Recommended prospective studies for validation
S. Samargandy et al., 2024	Partial - acknowledged single-center limitations	Mentioned results may not apply to other centers	Recommended caution in generalizing
Bengi Balci et al., 2023	Partial - discussed higher malignancy rates in subgroups	Noted impact of microcarcinomas on results	Recommended individualized decision-making
K. Anwar et al., 2023	Partial - noted convenient sampling technique	Not explicitly discussed	Recommended clinical judgment in doubtful cases
S. Ahmadi et al., 2019a	Partial - suggested validation in different cohort	Not explicitly discussed	Implied awareness of limitations
R. E. Khalushi et al., 2020	Partial - noted surgical selection due to fear of malignancy	Not explicitly discussed	Not provided
I. Ahmadinejad et al., 2023	Not mentioned	Not discussed	Not provided
Thayse Lozovoy Madsen Barbosa et al., 2019	Not mentioned	Not discussed	Not provided
Orhan Asya et al., 2022	Not mentioned	Not discussed	Not provided
S. Ahmadi et al., 2019	Not mentioned	Not discussed	Not provided
Wei-bing Zhang et al., 2020	Not mentioned	Not discussed	Not provided
M. D. de Jong et al., 2022	Not mentioned	Not discussed	Not provided
Ahmed Alsibani et al., 2024	Not mentioned	Not discussed	Not provided
Luying Gao et al., 2019	Not mentioned	Not discussed	Not provided
K. Daniels et al., 2020	Not mentioned	Not discussed	Not provided
Yi Zheng et al., 2018	Not mentioned	Not discussed	Not provided
Yun Zhu et al., 2020	Not mentioned	Not discussed	Not provided
Vinay Raj Thattarakkal et al., 2021	Not mentioned	Not discussed	Not provided
Saad M. Alqahtani et al., 2022	Not mentioned	Not discussed	Not provided
Chi Fung Chia et al., 2023	Not mentioned	Not discussed	Not provided
Kyla Wright et al., 2022	Not mentioned	Not discussed	Not provided
M. R. Romero Ibarguengoitia et al., 2024	Not mentioned	Not discussed	Not provided
M. Paker et al., 2021	Not mentioned	Not discussed	Not provided

Only 2 of 25 studies (8%) explicitly acknowledged surgical selection bias . Five additional studies (20%) provided partial acknowledgment through discussion of study limitations or recommendations for validation in different populations . The remaining 18 studies (72%) did not address selection bias in any form.

Impact on Reported Malignancy Prevalence

The heterogeneity in malignancy prevalence (12.2-66.1%) likely reflects the impact of surgical selection bias. Studies with higher malignancy rates tended to have more restrictive inclusion criteria:

- Studies limited to indeterminate cytology reported rates of 47.1-47.3%
- Studies with broader surgical indications reported rates of 22-50%
- Prospective consecutive enrollment studies reported the lowest rates (12.2-12.5%)

One study explicitly noted that observed malignancy risk was higher than expected for ACR TI-RADS categories 3-5 and ATA's very low, low, and intermediate suspicion categories . Similarly, another study observed that malignancy prevalence in their sample was lower than a German reference sample across all TI-RADS categories , suggesting geographic and referral pattern variations.

Synthesis

The substantial heterogeneity in reported malignancy prevalence and diagnostic performance across studies can be explained through several mechanisms rather than representing contradictory findings.

Population and Selection Criteria Effects

Studies with restrictive inclusion criteria (e.g., only indeterminate cytology or only TI-RADS category 5 nodules) consistently showed higher malignancy rates (47-66%) compared to studies with broader enrollment (12-27%) . The prospective studies with consecutive enrollment demonstrated the lowest malignancy rates (12.2-12.5%) , most closely approximating expected screening population rates. This gradient supports the conclusion that surgical cohort composition is the primary driver of reported malignancy prevalence, not differences in ACR TI-RADS performance.

Methodological Quality Considerations

The three largest studies (n>1000 nodules) provide the most stable estimates of category-specific malignancy rates, showing consistent gradients from TR1-2 (0-1.3%) through TR5 (67.1-88.8%) . Smaller surgical cohorts showed greater variability, particularly in intermediate categories where the AUS/FLUS-focused study found no significant association between TI-RADS and pathological outcome . Studies with explicit methodology for blinded TI-RADS assignment generally reported performance metrics consistent with larger validation cohorts.

Context-Specific Performance

ACR TI-RADS performance appears to vary by clinical context:

- In general surgical cohorts: Sensitivity 74-84%, specificity 53-77%
- In indeterminate cytology cohorts: Lower performance with accuracy ~50% and poor inter-reader reliability (κ=0.293)
- By nodule size: Better performance in nodules >1cm compared to ≤1cm

These findings suggest that ACR TI-RADS maintains good discriminatory ability in general surgical populations but may have limited additional value in pre-selected high-risk populations where cytology is already indeterminate.

Implications for Clinical Practice

The observed malignancy rates in surgical cohorts substantially exceed those expected for screening populations based on ACR TI-RADS thresholds. For TR3 nodules, surgical cohorts reported malignancy rates of 9-43% versus the expected ~5% in screening populations. For TR4 nodules, rates ranged from 13-60% versus expected 5-20%. This inflation reflects appropriate clinical decision-making whereby surgery is performed on nodules with additional concerning features beyond ultrasound appearance alone. The category-specific malignancy rates reported in surgical cohorts should not be used to estimate risk in screening populations, a limitation explicitly acknowledged by only 8% of included studies .

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